

TESTED & VERIFIED

UBUNTU

RDk X5

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A7672S 4G LTE Module RDk X5 Setup Guide

Complete Internet Setup — Vi, Airtel, BSNL & Jio
PPP Connection with Auto-Start on Boot

A comprehensive, step-by-step guide to connect an A7672S 4G LTE Cat 1 module to an RDk X5 board running Ubuntu. Covers hardware setup, AT command testing, PPP configuration, ECM conflict resolution, boot-time auto-start service, and troubleshooting. Covers all major Indian SIM operators.

A7672S LTE Cat 1

RDk X5

Ubuntu Linux

PPP / pppd

systemd Auto-Start

AT Commands

Minicom

SIM COMPATIBILITY AT A GLANCE

- | | |
|---|---|
|  Vi (Vodafone-Idea) — Tested & Working |  Airtel — Supported |
|  BSNL — Supported |  Jio — NOT SUPPORTED (VoLTE issue) |

Platform: RDK X5 (Ubuntu) | Module: A7672S Cat 4 LTE | Auto-Start on Boot ✓ | Tested Latency:
~47 ms

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Introduction & Overview

This guide provides a **complete, tested, and verified** setup to connect an **A7672S 4G LTE Cat 1 module** to an **RDK X5 board running Ubuntu** for automatic 4G internet access using SIM card data. The setup uses **PPP (Point-to-Point Protocol)** over a USB serial connection and auto-starts on every system boot — no manual commands needed.

This document was built from a real troubleshooting session that covered every edge case: wrong port detection, ECM/PPP conflicts, module hang recovery, SIM compatibility issues, and systemd service tuning. The result is a reliable, production-ready setup.

What You Will Achieve

- A7672S module connected to RDK X5 via USB
- Verified AT command communication with the module
- PPP internet connection configured for your SIM operator
- ECM/USB-network conflict permanently resolved
- Auto-start service: internet connects on every boot — zero commands needed
- Wi-Fi and 4G can run simultaneously without conflict



Tested Result

On RDK X5 with Ubuntu and a Vi 4G SIM — after reboot, the service connected automatically with IP 100.72.82.218, 0% packet loss, and ~47 ms latency to 8.8.8.8. No commands were typed.

How to Use This Guide

Follow the numbered steps in order. Each step builds on the previous one. If you encounter issues, refer to **Section 15 — Troubleshooting** for specific error messages and fixes. The **Quick Reference** in Section 16 provides a compact command card for future use.

SIM Card Compatibility



Jio SIM Does NOT Work with A7672S — Do Not Use!

- Jio operates as a **VoLTE-only** network in India.
- The A7672S is a **Cat 1 LTE module** without full VoLTE/IMS support.
- Jio's network sends a **Registration DENIED** response (AT+CREG? returns +CREG: 0,3 instead of 0,1).
- Even if AT commands respond, **data sessions cannot be activated**: AT+CGATT=1 returns +CME ERROR: operation not allowed.
- Switching to a different APN or chat script will NOT fix this — it is a network-level rejection.

— **Bottom line: Use Vi (Vodafone-Idea), Airtel, or BSNL. Jio SIM will never work on this module regardless of configuration.**

Supported SIM Operators

OPERATOR	APN	STATUS	NOTES
Vi (Vodafone-Idea)	vi-internet	✓ Tested & Working	Best compatibility; fully tested in this guide
Airtel	airtelgprs.com	✓ Supported	Works well with Cat 1 LTE modules
BSNL	bsnlnet	✓ Supported	4G coverage may vary by region
Jio	jionet	✗ NOT Supported	VoLTE-only; A7672S Cat 1 is rejected by network (CREG: 0,3)



Why Does Jio Fail?

Jio requires IMS/VoLTE registration before allowing any data bearer. The A7672S Cat 1 module does not perform a full IMS registration, so Jio's network denies the device entirely — it is not an APN, SIM, or configuration problem. This is a known limitation of Cat 1 modules on Jio India.

SECTION 3

Hardware Requirements

COMPONENT	DETAILS	REQUIRED?
A7672S 4G LTE Module	Cat 1 LTE, SIM slot, USB interface	Required
RDK X5 Board	Running Ubuntu Linux, USB ports available	Required
SIM Card	Vi, Airtel, or BSNL with active 4G data plan	Required
USB Cable	Connect A7672S USB port to RDK X5 USB port	Required
LTE Antenna	4G/LTE antenna attached to A7672S module	Required
Internet (for apt)	Wi-Fi or Ethernet for initial package install	Recommended



Power Supply Note

The A7672S module can draw up to ~2A peak current during transmission. Ensure the USB port on RDK X5 can supply adequate current. If the module's LED stops blinking or the module crashes frequently, use a powered USB hub or an external 5V/2A supply for the module.

SECTION 4

Connection Method

USB Connection (Recommended)

The simplest and most reliable method. The A7672S module connects via USB to the RDK X5 board. Ubuntu will automatically detect it as a USB serial device and create `/dev/ttyUSB*` ports.

- 1 Insert your SIM card into the A7672S module slot.
- 2 Attach the LTE antenna to the module's antenna connector.
- 3 Connect the A7672S module to any USB port on the RDK X5 board using a USB cable.
- 4 Wait 15–20 seconds for the module to boot. The LED should start blinking (3-second interval indicates SIM connected).



LED Status Guide

Slow blink (3s interval): Module registered on network with SIM 

Fast blink: Searching for network

Constant ON: Module powered but not yet registered

No LED: Module not powered or crashed — unplug USB and replug

Alternative: UART / GPIO Connection

For advanced use, connect via the RDK X5's 40-pin GPIO header (UART):

A7672S PIN	RDK X5 GPIO PIN	SIGNAL
TX	RX (UART)	Serial Data
RX	TX (UART)	Serial Data
GND	GND	Ground
VCC	5V / External	Power (~4V needed)

Note: UART method requires separate power for the module. USB method is simpler and recommended.

Device Detection & Identifying the AT Port

After connecting the module via USB, verify Ubuntu has detected it:

```
TERMINAL

# List USB serial ports – should show 4 ports
ls /dev/ttyUSB*

# Expected output:
/dev/ttyUSB0 /dev/ttyUSB1 /dev/ttyUSB2 /dev/ttyUSB3

# Also verify USB device is detected
lsusb

# Look for "SIMCom" or vendor ID "1e0e" in the output
```

Finding the Correct AT Command Port

The A7672S exposes 4 USB serial ports. Only one of them responds to AT commands. **Run this loop to find it automatically:**

```
PORT DETECTION LOOP — RUN IN TERMINAL

for port in 0 1 2 3; do
  echo "--- Testing ttyUSB$port ---"
  sudo sh -c "echo -e 'AT
' > /dev/ttyUSB$port"
  sleep 1
  sudo timeout 2 cat /dev/ttyUSB$port
  echo ""
done
```



How to Read the Output

The port that prints **"OK"** is your AT command port. In most A7672S setups this is /dev/ttyUSB1 or /dev/ttyUSB2. Note this port — you will use it in all subsequent steps.



Port Assignment Can Change After USB Replug

If you unplug and replug the USB cable, the port numbers may change (e.g., previously ttyUSB2 may become ttyUSB1). Always re-run the detection loop after a physical USB reset.

Install Required Packages

Ensure the following packages are installed on your Ubuntu system:

TERMINAL

```
sudo apt update  
sudo apt install minicom ppp net-tools -y
```

PACKAGE	PURPOSE
minicom	Serial terminal — for sending AT commands and testing the module
ppp	PPP daemon — creates the ppp0 network interface and manages the connection
net-tools	Provides ifconfig command to check network interface status

AT Command Testing & SIM Verification

Before configuring PPP, verify the module and SIM are working correctly using Minicom (serial terminal).

OPEN MINICOM (REPLACE TTYUSB1 WITH YOUR DETECTED PORT)

```
sudo minicom -D /dev/ttyUSB1 -b 115200
```

If keyboard input is not working inside minicom:

Press Ctrl+A → E (to toggle echo ON)

To exit minicom: Ctrl+A → X → press Enter

Required AT Commands — Run in This Order

COMMAND	EXPECTED RESPONSE	MEANING	STATUS
AT	OK	Module is responding	✓ Must pass
AT+CPIN?	+CPIN: READY	SIM card detected and unlocked	✓ Must pass
AT+CSQ	+CSQ: 22,99	Signal strength (10–31 = good; 0–9 = weak)	✓ Should be ≥ 10
AT+CREG?	+CREG: 0,1	Registered on home network	✓ Must be 0,1
AT+CGREG?	+CGREG: 0,1	Data network (GPRS/LTE) registered	✓ Must be 0,1

CREG Response Code Reference

RESPONSE	MEANING	ACTION REQUIRED
+CREG: 0,0	Not registered, not searching	Check SIM and signal
+CREG: 0,1	✓ Registered — Home network	Proceed to next step
+CREG: 0,2	Searching for network...	Wait and retry
+CREG: 0,3	✗ Registration DENIED	SIM not compatible — switch to Vi/Airtel/BSNL
+CREG: 0,5	Registered — Roaming	Will work but roaming charges may apply

✓ ACTUAL TEST RESULTS — VI SIM ON A7672S

AT+CPIN?
+CPIN: READY

AT+CSQ
+CSQ: 22,99 ← Signal: 22/31 (Good)

AT+CREG?
+CREG: 0,1 ← Registered on home network ✓

AT+CGREG?
+CGREG: 0,1 ← Data network registered ✓



If CREG Returns 0,3 — Stop Here

If AT+CREG? returns +CREG: 0,3, your SIM is being rejected by the network. **This is the Jio problem.** No amount of configuration changes will fix it. Replace the SIM with Vi, Airtel, or BSNL before continuing.

Exit Minicom after testing: Press Ctrl+A then X, then Enter. **Never leave Minicom open** when starting PPP — it will lock the port.

Chat Script Configuration

The chat script tells the PPP daemon how to dial the network. **Only the APN line changes between operators.** Everything else is identical.

CREATE / EDIT THE CHAT SCRIPT

```
sudo nano /etc/chatscripts/a7672s
```

Vi (Vodafone-Idea)

```
APN: VI-INTERNET

ABORT 'BUSY'
ABORT 'NO CARRIER'
ABORT 'ERROR'
TIMEOUT 30
'' AT
OK ATE0
OK AT+CGDCONT=1,"IP","vi-internet"
OK ATD*99#
CONNECT ''
```

Airtel

```
APN: AIRTELGPRS.COM

ABORT 'BUSY'
ABORT 'NO CARRIER'
ABORT 'ERROR'
TIMEOUT 30
'' AT
OK ATE0
OK AT+CGDCONT=1,"IP","airtelgprs.com"
OK ATD*99#
CONNECT ''
```

BSNL

```
APN: BSNLNET

ABORT 'BUSY'
ABORT 'NO CARRIER'
ABORT 'ERROR'
TIMEOUT 30
'' AT
OK ATE0
OK AT+CGDCONT=1,"IP","bsnlnet"
OK ATD*99#
CONNECT ''
```

Jio — Not Supported

```
 DO NOT USE — WILL NEVER WORK

ABORT 'BUSY'
ABORT 'NO CARRIER'
ABORT 'ERROR'
TIMEOUT 30
'' AT
OK ATE0
OK AT+CGDCONT=1,"IP","jionet"
OK ATD*99#
CONNECT ''
```

LINE	WHAT IT DOES	CHANGE FOR DIFFERENT SIM?
ABORT 'BUSY'	Abort if line is busy	No
ABORT 'ERROR'	Abort on error response	No
TIMEOUT 30	Wait 30 sec for each response	No
' ' AT	Wake up the module	No
OK ATE0	Disable echo	No
OK AT+CGDCONT=1,"IP","APN"	Set carrier APN — this is the only line you change	Yes — APN only
OK ATD*99#	Dial the data connection	No
CONNECT ' '	Connection established signal	No

Save: Ctrl+O → Enter → Ctrl+X

SECTION 9 — STEP 5

PPP Peer File Configuration

The peer file tells pppd which serial port to use and how to handle the connection. **This file is the same for all SIM operators** — only the port may differ.

CREATE / EDIT PEER FILE

```
sudo nano /etc/ppp/peers/a7672s
```

CONTENT — SAME FOR ALL SIM OPERATORS

```
/dev/ttyUSB1
115200
connect '/usr/sbin/chat -v -f /etc/chatscripts/a7672s'
noauth
defaultroute
replacedefaultroute
usepeerdns
persist
```

OPTION	PURPOSE
<code>/dev/ttyUSB1</code>	AT command port (change to your detected port)
<code>115200</code>	Baud rate for serial communication
<code>noauth</code>	No local authentication required
<code>defaultroute</code>	Set ppp0 as the default route when connected
<code>replacedefaultroute</code>	Replace existing default route with ppp0
<code>usepeerdns</code>	Use DNS servers provided by the carrier
<code>persist</code>	Automatically reconnect if connection drops

Save: Ctrl+O → Enter → Ctrl+X

Manual Connection Test

Before setting up auto-start, verify the connection works manually. **Ensure Minicom is closed** before running these commands.

CONNECT & TEST

```
# Start PPP connection
sudo pon a7672s

# Wait 15 seconds for connection to establish
sleep 15

# Check ppp0 interface – should show an IP address
ifconfig ppp0

# Test internet connectivity
ping -c 4 8.8.8.8
```

✓ EXPECTED SUCCESSFUL OUTPUT

```
ppp0: flags=4305<UP,POINTOPOINT,RUNNING,NOARP,MULTICAST> mtu 1500
    inet 100.72.82.218 netmask 255.255.255.255 destination 10.64.64.64
    ppp txqueuelen 3 (Point-to-Point Protocol)
    RX packets 11 bytes 776 (776.0 B)
    TX packets 21 bytes 1247 (1.2 KB)

PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=115 time=41.9 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=115 time=41.7 ms

--- 8.8.8.8 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss ✓
```

If Manual Test Succeeds

Great! Disconnect and proceed to Step 7 (ECM Blacklist) then Step 8 (Auto-Start Service).

```
# Disconnect PPP
sudo poff a7672s
```

If Manual Test Fails

- Check CREG = 0,1 (not 0,3)
- Verify AT port in peer file is correct
- Kill stale processes: `sudo killall pppd 2>/dev/null`
- Remove lock: `sudo rm -f /var/lock/LCK..ttyUSB*`
- Check SIM has active data plan

ECM Interface Blacklist — CRITICAL Step



This Step is Mandatory — Do Not Skip!

The A7672S module creates an additional USB network interface (enxae0c29a39b6d or similar) via the **ECM (Ethernet Control Model)** driver. This interface conflicts with PPP — when both are active, PPP connects and authenticates but IPCP times out, meaning **no IP address is ever assigned**. The ECM driver must be blacklisted to allow PPP to work correctly.

CREATE THE BLACKLIST FILE

```
sudo nano /etc/modprobe.d/blacklist-ecm.conf
```

FILE CONTENT

```
blacklist cdc_ether
blacklist rndis_host
```

Save: Ctrl+O → Enter → Ctrl+X

DRIVER	WHAT IT CREATES	WHY BLACKLIST IT
<code>cdc_ether</code>	USB ECM network interface (enx...)	Conflicts with PPP; module cannot handle both simultaneously
<code>rndis_host</code>	RNDIS USB network interface	Same conflict; preventively disabled



This Takes Effect After Reboot

The blacklist is applied at boot time when kernel modules load. After creating this file, **reboot the system** for it to take effect. The enx... interface will no longer appear in `ifconfig` output, and PPP will be able to obtain an IP address from the carrier.



Also Remove the ECM Line from Service File

If you had previously added `ExecStartPre=/sbin/ifconfig enx...` down to the service file, **remove that line** now — the interface no longer exists, and that command will cause the service to fail with exit code 255.

Auto-Start Service — Internet on Every Boot

Create a systemd service that automatically starts the 4G internet connection every time the RDK X5 boots. No manual commands will be needed after this step.

CREATE THE SERVICE FILE

```
sudo nano /etc/systemd/system/a7672s-internet.service
```

SERVICE FILE CONTENT (FINAL, TESTED VERSION)

```
[Unit]
Description=A7672S 4G LTE Internet Connection
After=network.target

[Service]
Type=simple
ExecStartPre=/bin/sleep 20
ExecStart=/usr/sbin/pppd call a7672s nodetach
Restart=on-failure
RestartSec=30
TimeoutStartSec=120

[Install]
WantedBy=multi-user.target
```

PARAMETER	VALUE	WHY
<code>ExecStartPre=/bin/sleep 20</code>	Wait 20 seconds before starting	Module needs time to boot, register on network, and stabilize after power-on
<code>nodetach</code>	Keep pppd in foreground	Allows systemd to track the process correctly with <code>Type=simple</code>
<code>Restart=on-failure</code>	Auto-restart on failure	Recovers from temporary disconnections or dial failures
<code>RestartSec=30</code>	Wait 30 sec before retry	Prevents rapid retries that can hang the module; allows it to recover
<code>TimeoutStartSec=120</code>	Allow 2 minutes to connect	Prevents systemd from killing the service before connection is established
<code>persist</code>	In peer file (not here)	PPP-level auto-reconnect if carrier drops the connection

Enable and Apply

RUN THESE COMMANDS AFTER SAVING THE SERVICE FILE

```
# Reload systemd to recognize the new service file
sudo systemctl daemon-reload

# Enable auto-start on boot
sudo systemctl enable a7672s-internet.service

# Reboot to apply the ECM blacklist AND test the service
sudo reboot
```

Service Management Commands

TASK	COMMAND
Start internet now	<code>sudo systemctl start a7672s-internet.service</code>
Stop internet	<code>sudo systemctl stop a7672s-internet.service</code>
Restart connection	<code>sudo systemctl restart a7672s-internet.service</code>
Check status	<code>sudo systemctl status a7672s-internet.service</code>
View logs	<code>sudo journalctl -xeu a7672s-internet.service</code>
Disable auto-start	<code>sudo systemctl disable a7672s-internet.service</code>

Verification After Reboot

After rebooting, wait at least **60 seconds** for the module to boot, register on the network (20 seconds sleep), and establish the PPP connection. Then run the following checks:

PASTE ALL AT ONCE IN TERMINAL

```
sleep 60
sudo systemctl status a7672s-internet.service
ifconfig ppp0
ping -c 4 8.8.8.8
```

What to Look For

CHECK	EXPECTED	IF DIFFERENT
Service Active line	Active: active (running)	See Troubleshooting Section 15
ppp0 flags	UP, POINTOPOINT, RUNNING	Connection not established yet — wait more
ppp0 inet address	100.x.x.x or 10.x.x.x	IPCP failed — check ECM blacklist
Ping packet loss	0% packet loss	Data routing issue; check default route

✓ VERIFIED BOOT RESULT — VI SIM ON RDK X5

- a7672s-internet.service - A7672S 4G LTE Internet Connection
Active: active (running) since Thu 2026-06-11 17:38:37 CST; 3min 43s ago
Process: ExecStartPre=/bin/sleep 20 (status=0/SUCCESS)
Main PID: 1318 (pppd)

Jun 11 17:38:42 ubuntu pppd[1318]: local IP address 100.72.82.218

Jun 11 17:38:42 ubuntu pppd[1318]: remote IP address 10.64.64.64

Jun 11 17:38:42 ubuntu pppd[1318]: primary DNS address 10.216.215.103

ppp0: flags=4305<UP,POINTOPOINT,RUNNING,NOARP,MULTICAST> mtu 1500

inet 100.72.82.218 netmask 255.255.255.255

RX packets 819 bytes 561624 ✓ Data flowing!

4 packets transmitted, 4 received, 0% packet loss, ~47 ms ✓



Setup Complete!

Your RDK X5 now automatically connects to the internet via the A7672S 4G module on every boot — no commands needed. Wi-Fi and 4G can run simultaneously without conflict.

SECTION 14

Changing SIM Card Operator

When switching to a different SIM operator, **only one file needs to change** — the APN in the chat script. Everything else remains identical.

- 1 Power off module: unplug the USB cable from RDK X5.
- 2 Physically swap the SIM card in the A7672S module.
- 3 Edit the chat script and change only the APN line:
`sudo nano /etc/chatscripts/a7672s`
- 4 Plug USB back in and reboot: `sudo reboot`

NEW SIM OPERATOR	CHANGE THE APN LINE TO	WORKS WITH A7672S?
Vi (Vodafone-Idea)	OK AT+CGDCONT=1,"IP","vi-internet"	✓ Tested & Working
Airtel	OK AT+CGDCONT=1,"IP","airtelgprs.com"	✓ Supported
BSNL	OK AT+CGDCONT=1,"IP","bsnlnet"	✓ Supported
Jio	OK AT+CGDCONT=1,"IP","jionet"	✗ NOT Supported — Do Not Use



Jio SIM: CREG 0,3 — Permanent, Unfixable Issue

- Symptom: AT+CREG? returns +CREG: 0,3 (Registration DENIED)
- AT+CGATT=1 returns +CME ERROR: operation not allowed
- No APN, script, or driver change will fix this
- This is a Jio network-level rejection of non-VoLTE Cat 1 modules



Solution: Replace Jio SIM with Vi, Airtel, or BSNL.

PROBLEM / SYMPTOM	LIKELY CAUSE	SOLUTION
No /dev/ttyUSB* ports	Module not detected	Check USB cable; run <code>lsusb</code> ; try different USB port
AT commands not responding	Wrong port, or module hung	Run port detection loop; USB unplug 10 sec → replug → wait 15 sec
Module doesn't respond on <i>any</i> port	Module hung/crashed (common after many failed retries)	Physical reset only: USB unplug → 10 sec → replug → 15 sec wait → LED should blink
Device <code>ttyUSB1</code> is locked by pid ...	Previous <code>pppd</code> process still running	<code>sudo killall pppd</code> then <code>sudo rm -f /var/lock/LCK..ttyUSB*</code>
IPCP timeout / no IP address on <code>ppp0</code>	ECM interface conflict	Ensure <code>/etc/modprobe.d/blacklist-ecm.conf</code> exists with <code>cdc_ether</code> blacklisted; reboot
CREG: 0,3 — Registration DENIED	Incompatible SIM (Jio)	Switch to Vi / Airtel / BSNL SIM
Service keeps restarting in loop	Module not ready at boot; wrong port	Increase sleep: change <code>sleep 20</code> to <code>sleep 30</code> in service file
Service fails with exit code 255	ExecStartPre line referencing ECM interface that no longer exists	Remove the <code>ifconfig enx...</code> down line from service file
<code>ppp0</code> has no IP after service starts	IPCP timeout (carrier not responding)	Verify SIM has active data plan; check signal with <code>AT+CSQ</code>
<code>poff a7672s</code> fails / takes long	<code>pppd</code> stuck	<code>sudo killall pppd</code> and <code>sudo rm -f /var/lock/LCK..ttyUSB*</code>
Internet works but DNS fails	DNS not configured	<code>echo "nameserver 8.8.8.8" sudo tee /etc/resolv.conf</code>
Module LED stops / goes constant ON	Power issue or crash	USB unplug-replug; check power supply can provide ≥2A
AT port changed after USB replug	Linux re-assigned port numbers	Re-run port detection loop; update <code>/etc/ppp/peers/a7672s</code> with correct port

Module Physical Reset Procedure

When the module becomes completely unresponsive (no AT response on any port after all detection attempts):

- 1 Stop the service: `sudo systemctl stop a7672s-internet.service`
- 2 Kill any pppd: `sudo killall pppd 2>/dev/null && sudo rm -f /var/lock/LCK..ttyUSB*`
- 3 Physically unplug the USB cable from RDK X5.
- 4 Wait 10 full seconds.
- 5 Plug the USB cable back in.
- 6 Wait 15–20 seconds for module to boot (LED should start blinking).
- 7 Verify ports: `ls /dev/ttyUSB*` then re-run detection loop.

Quick Reference Command Card

Everyday Commands

TASK	COMMAND
Start internet	<code>sudo systemctl start a7672s-internet.service</code>
Stop internet	<code>sudo systemctl stop a7672s-internet.service</code>
Restart connection	<code>sudo systemctl restart a7672s-internet.service</code>
Check status	<code>sudo systemctl status a7672s-internet.service</code>
Check ppp0	<code>ifconfig ppp0</code>
Test connectivity	<code>ping -c 4 8.8.8.8</code>
View logs	<code>sudo journalctl -xeu a7672s-internet.service</code>

Troubleshooting Commands

TASK	COMMAND
Manual connect	<code>sudo pon a7672s</code>
Manual disconnect	<code>sudo poff a7672s</code>
Kill stuck pppd	<code>sudo killall pppd 2>/dev/null</code>
Remove lock file	<code>sudo rm -f /var/lock/LCK..ttyUSB*</code>
Check signal	Minicom → AT+CSQ
Check registration	Minicom → AT+CREG?
Find AT port	Run port detection loop

Key File Locations

FILE	PURPOSE	WHAT TO CHANGE
<code>/etc/chatscripts/a7672s</code>	Dial script with APN	APN only (when switching operators)
<code>/etc/ppp/peers/a7672s</code>	Port, speed, PPP options	Port (ttyUSB1, etc.) if it changes after replug
<code>/etc/systemd/system/a7672s-internet.service</code>	Auto-start service definition	Sleep time if module needs more boot time
<code>/etc/modprobe.d/blacklist-ecm.conf</code>	Disables ECM/RNDIS USB drivers	Do not modify

SECTION 17

Module Specifications

SPECIFICATION	VALUE	SPECIFICATION	VALUE
Module Model	A7672S	Connection Method	PPP over USB serial
Category	LTE Cat 1	Tested Latency	~47 ms (Vi SIM)
Max Download Speed	10 Mbps	OS	Ubuntu (on RDK X5)
Max Upload Speed	5 Mbps	Auto-Start	Yes (systemd)
Supported Modes	LTE-FDD / GSM / GPRS / EDGE	Auto-Reconnect	Yes (persist + Restart)

SECTION 18

Important Notes & Best Practices

1. **Always use Vi, Airtel, or BSNL SIM** — Jio is permanently incompatible with A7672S (VoLTE-only network).
2. **ECM blacklist is mandatory** — Without `blacklist-ecm.conf`, PPP will never get an IP address (IPCP: timeout).
3. **sleep 20 is critical** — The module needs time after power-on to register on the cellular network before PPP is attempted.
4. **Physical USB reset is the only fix for hung module** — When the module stops responding to AT commands on all ports, only unplugging and replugging the USB cable (with 10 seconds gap) will recover it.
5. **Wi-Fi and 4G can run simultaneously** — No conflict between `wlan0` and `ppp0` when ECM is properly blacklisted.
6. **The `persist` option** in the peer file ensures PPP-level auto-reconnect if the carrier temporarily drops the connection.
7. **After USB replug, the AT port may change** — Always re-run the port detection loop and update the peer file if the port changes.
8. **Never run Minicom while the service is active** — Both compete for the same serial port and will lock each other out.

9. **RestartSec=30** prevents rapid-fire retries — rapid retries crash the module; 30 seconds gives it recovery time.
10. **Ensure SIM has an active data plan** — A depleted data balance causes the same symptoms as network issues (0% RX traffic, IPCP failures).

A7672S 4G LTE Module — RDK X5 Complete Setup Guide v1.0 | Tested on Ubuntu (RDK X5) with Vi 4G SIM | This guide covers Vi, Airtel, BSNL (supported) and Jio (not supported). For Jio: the module is fundamentally incompatible — VoLTE-only network rejects Cat 1 devices.